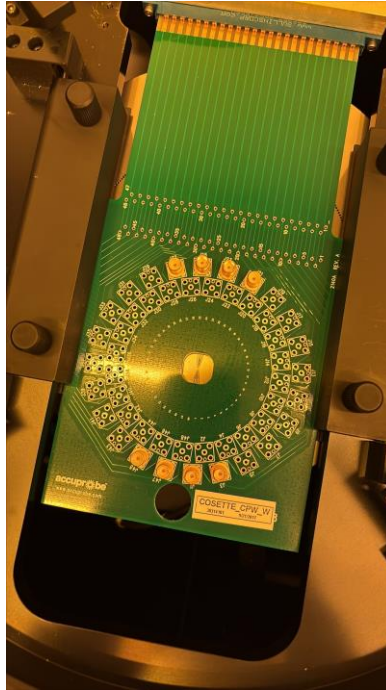


Probe08 CenFire Inline Test Procedure

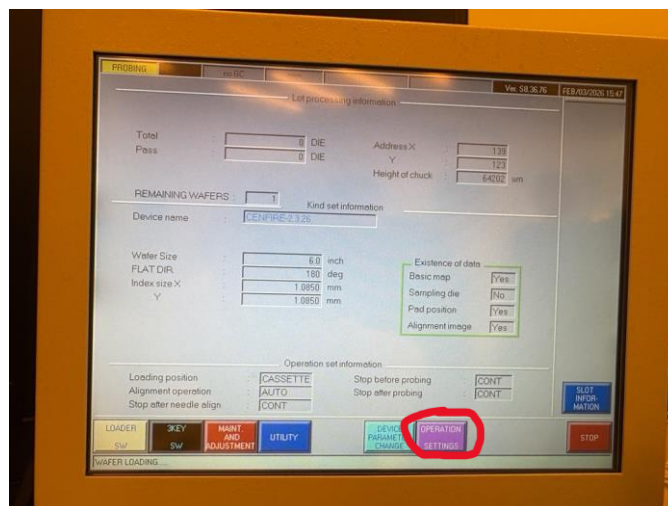
1. Install Probe Card:

- Before starting test, ensure that proper probe card is installed on PROBE8

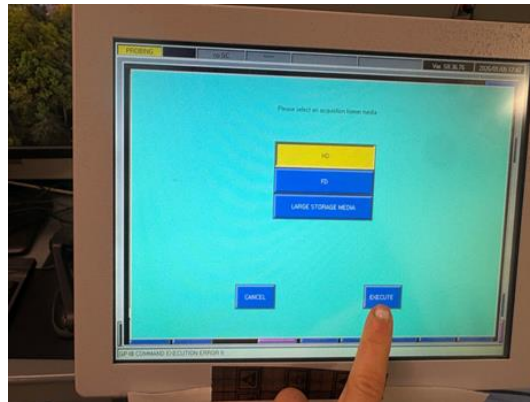


2. Configure Operator Settings

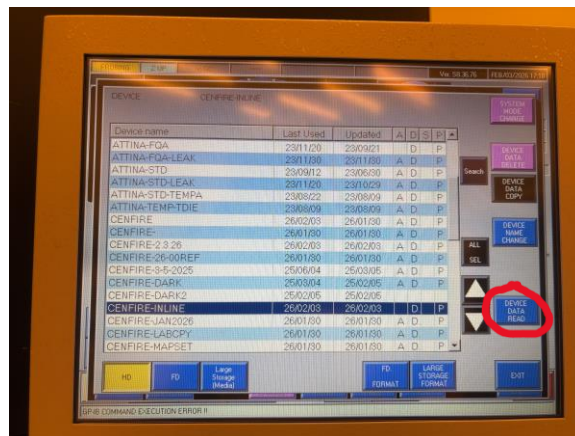
- Configure Operator settings:



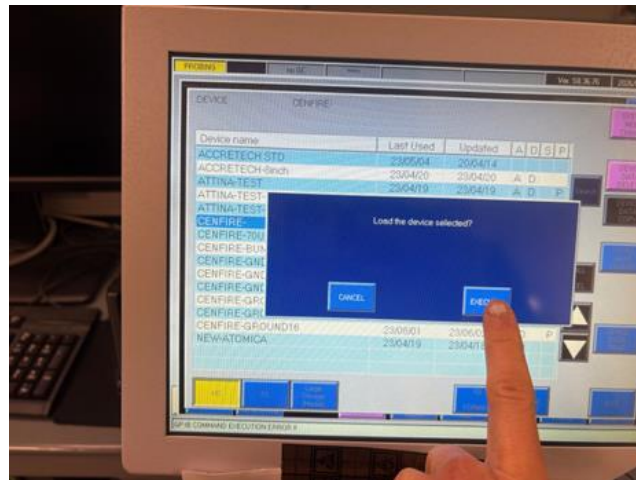
- Once operator Settings match the image, select "ENT"



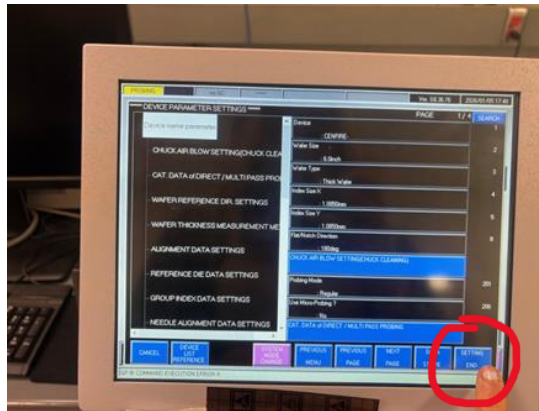
- d. Select the device: CENFIRE-INLINE from the options displayed and DEVICE DATA READ



- e. Select EXECUTE to load **CENFIRE-INLINE**

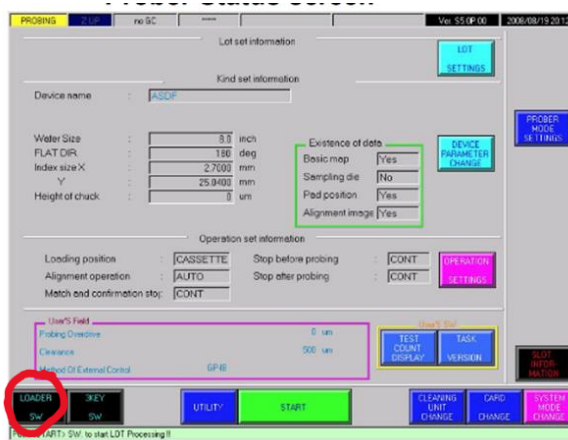


- f. Select SETTING END to return to the main screen

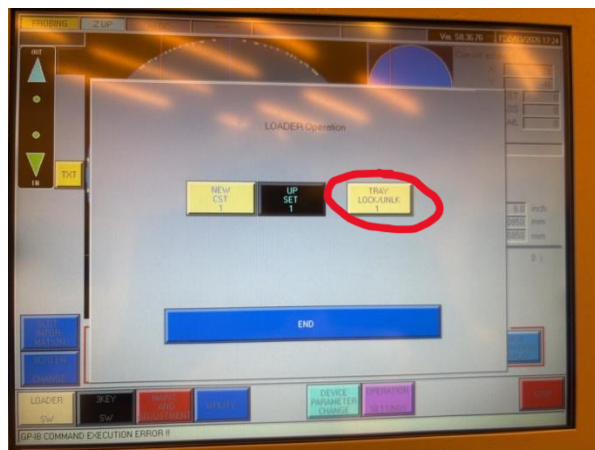


4. Load the Wafer

a. Select LOADER SW

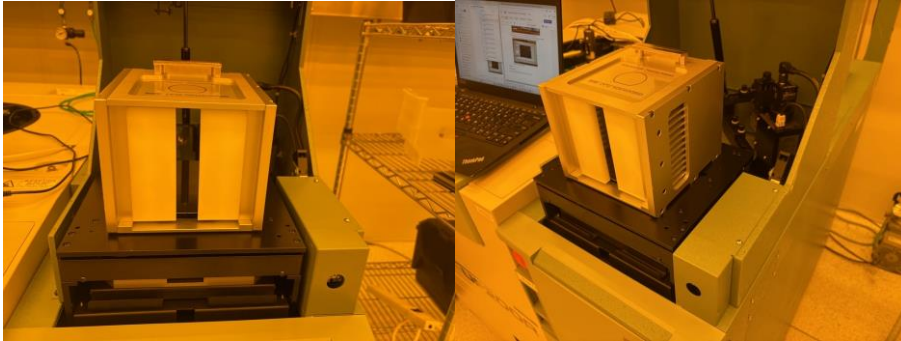


b. Select TRAY LOCK/UNLK to toggle between loader lock/unlock

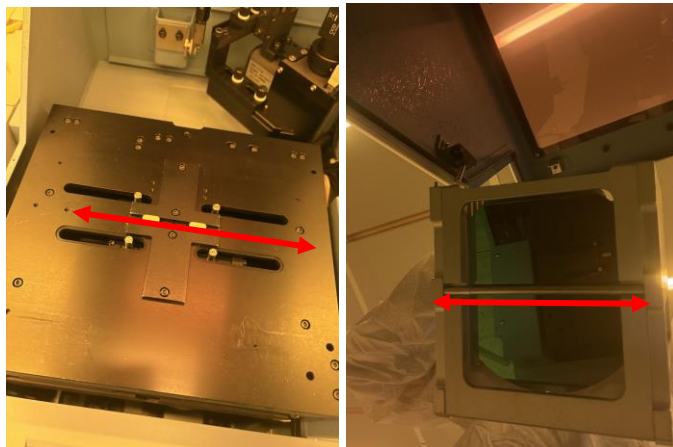


a. Load the wafer(s) in the cassette. Wafer should be inserted in the cassette in the LOADER, with the notch facing out (toward the operator).

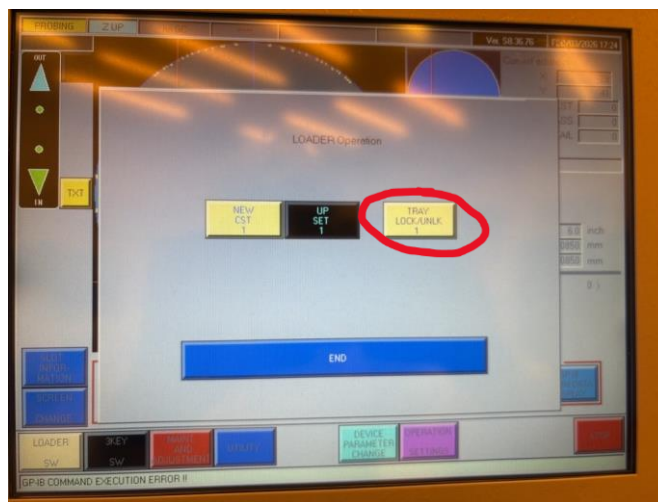
- a. Note: If wafers require manual alignment, the prober will stop for alignment between each wafer load
- b. When loading the wafers, track the wafer IDs in order from bottom to top row of cassette. You will need to record wafer ID in test program for each wafer.



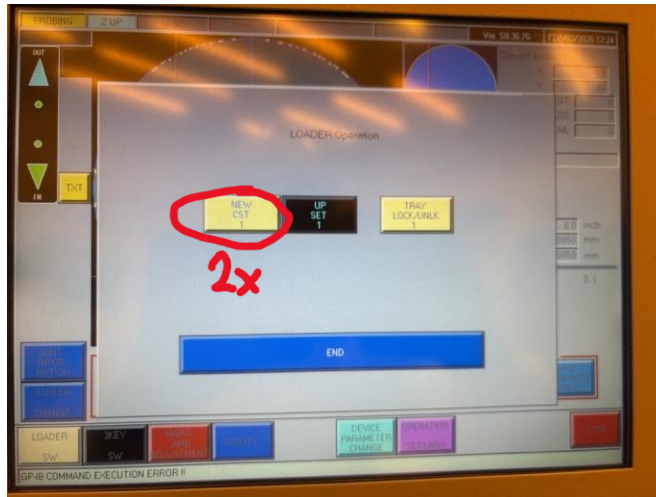
- c. When loading the cassette, make sure that the horizontal bar on bottom of cassette is slotted into horizontal slot in loader and cassette is centered in the loader:



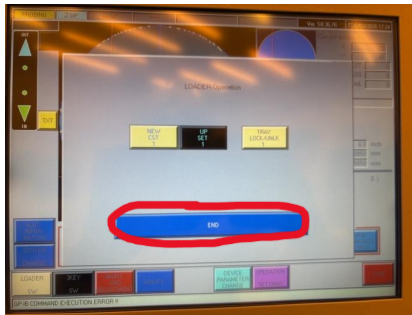
- d. Select TRAY LOCK/UNLK to lock the Loader



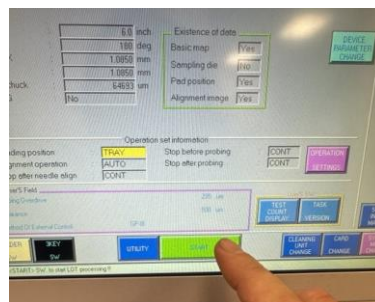
- e. Select NEW CST, cassette will lower. You may have to open LOADER SW and select NEW CST twice before it lowers.



- f. Select END to close window



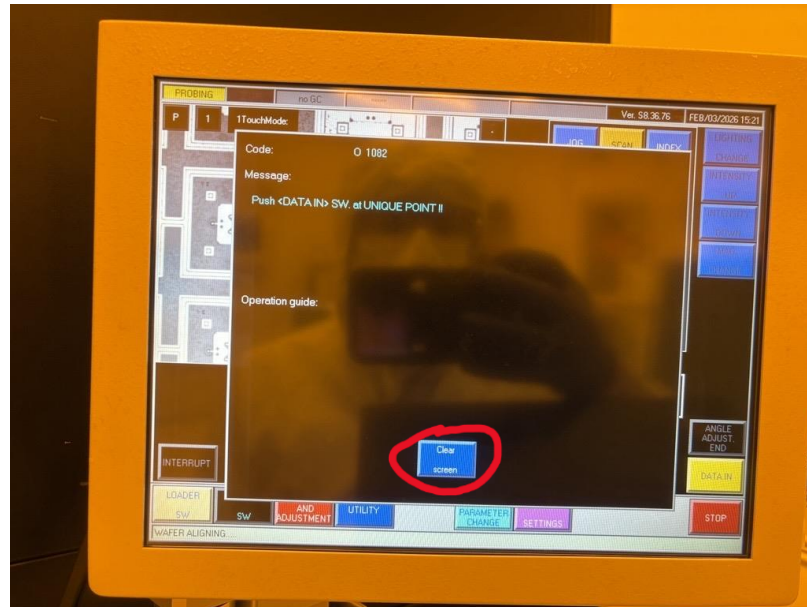
- g. Select START on the main window to start the loading process.



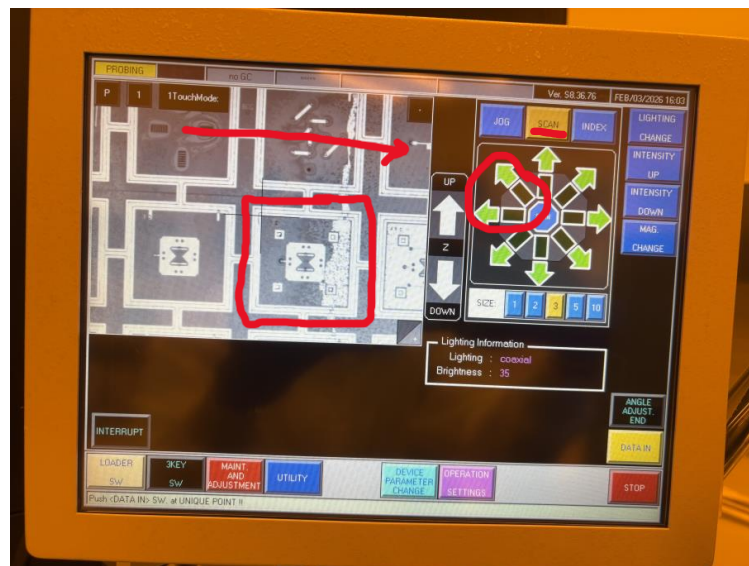
5. Wait for wafer to load (prober will start to prompt for user inputs)

6. Manually Align wafer

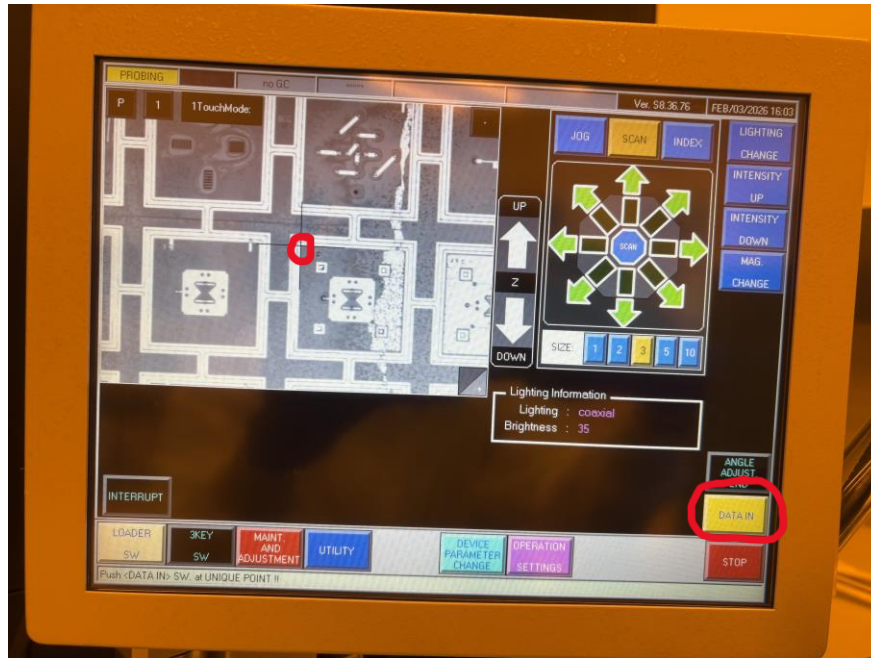
1. When prompted to find unique point, clear screen:



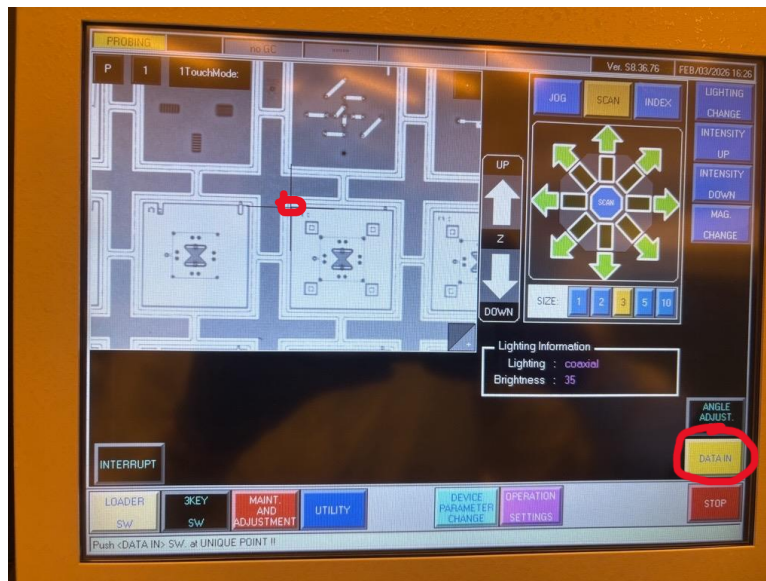
2. Using the SCAN function and pressing the green arrows to move, Find a die at the top of a reticle. This die has all test structures above it (test structures look different than product die)



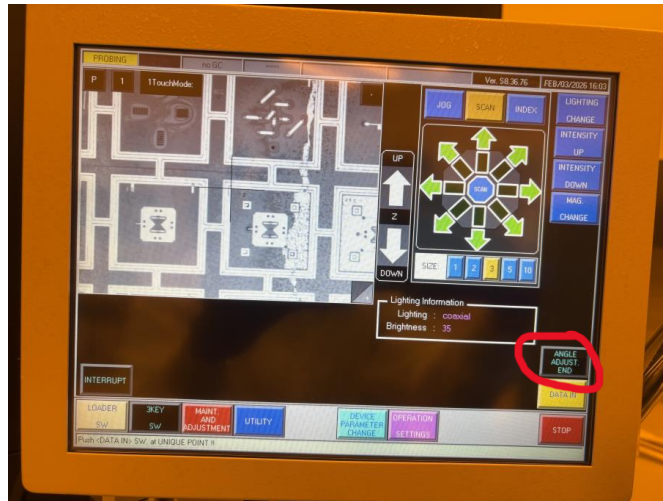
3. line up cross hair with top left inner corner of the die in the top row, then Select DATA IN



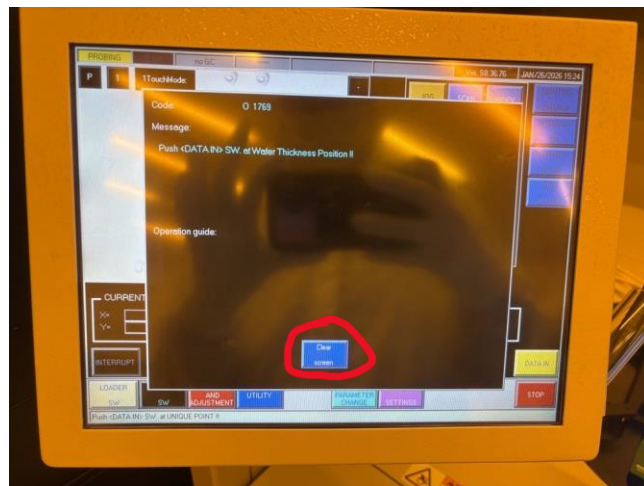
4. The camera will move. For chuck theta alignment, you must find the same point on a die in the same row as the first. Find a die in the top row of the reticle (test structures in the row above), and line up the crosshairs to the top left corner. Select DATA IN.



5. Camera will toggle between these unique points. It should only scan left to right, there should be no vertical motion.
 - a. To restart the alignment process, Hit INTERRUPT twice, then START
 - b. To unload wafer and try again, hit INTERRUPT twice, then UNLOAD
6. Once wafer is aligned and camera motion is only left to right (no vertical movement), hit "ANGLE ADJUST END"



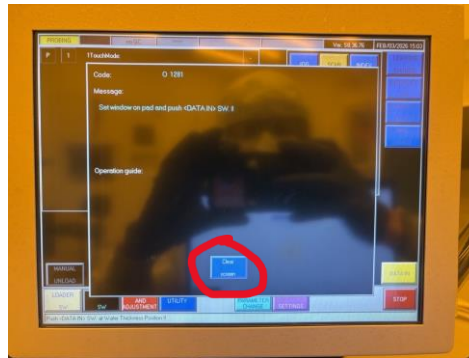
7. CLEAR SCREEN at Push Data in SW at Wafer Thickness Position



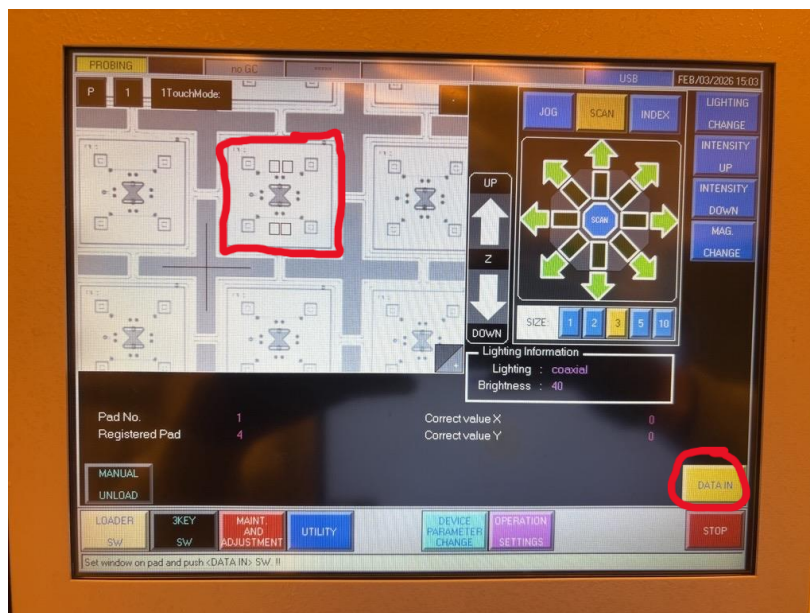
8. Use green SCAN arrows to move cross hairs next to die structure and select DATA IN
 - a. Use "INTENSITY DOWN/UP" to adjust lighting if needed



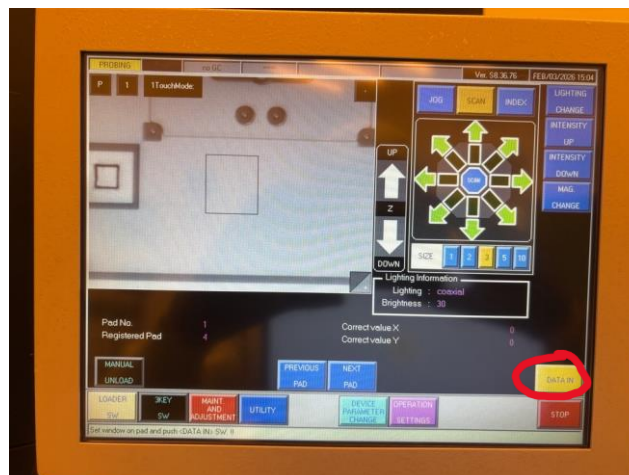
9. At Set Window on Pad prompt, clear screen



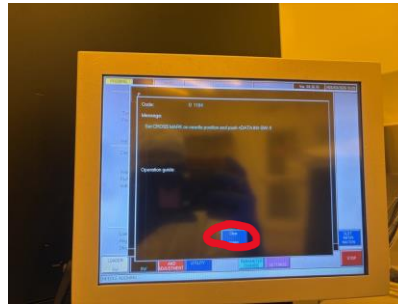
10. Pads will overlay on product die. Do not adjust. Hit DATA IN.



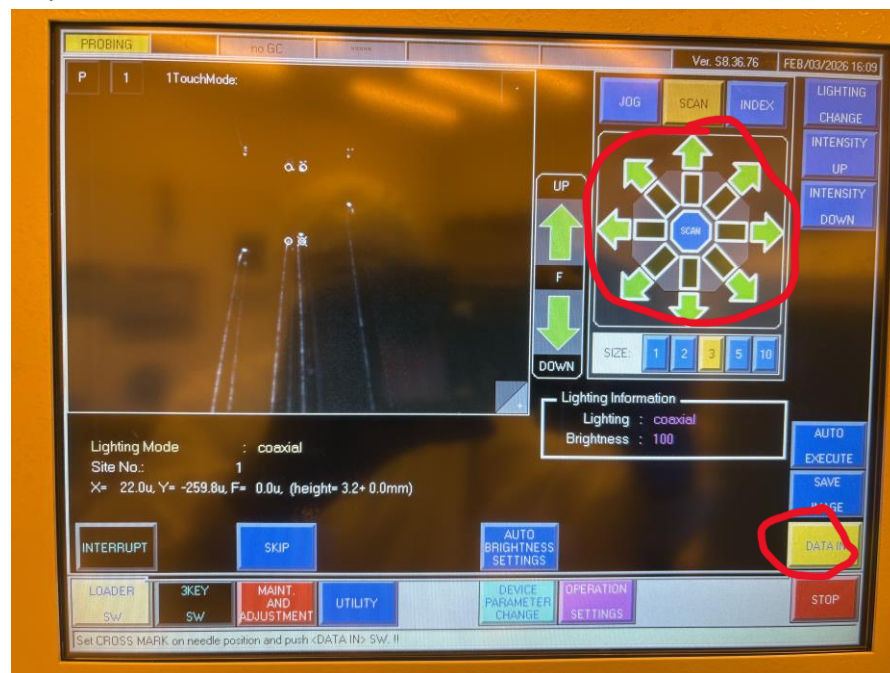
11. When zooms in on pad, do not adjust, hit DATA IN



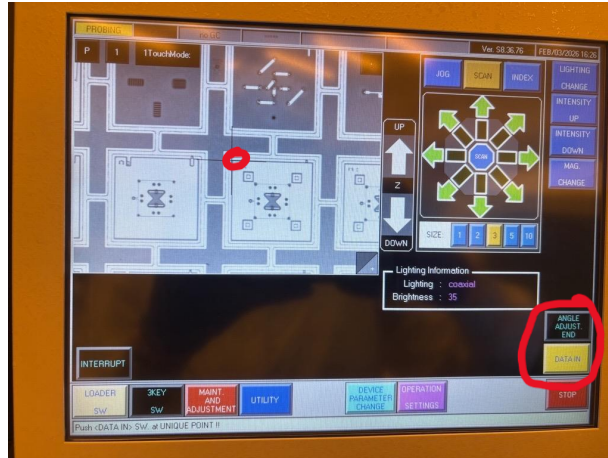
12. Clear Screen at the prompt to put the cross marks on the needle positions



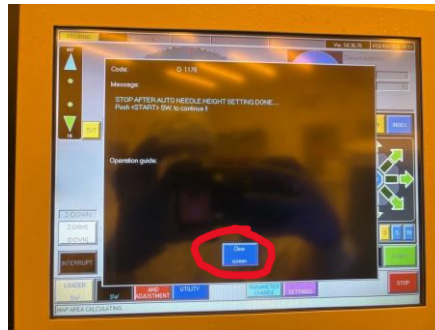
13. Use the green SCAN arrows to set the Cross marks as close as you can to the needle tips and hit DATA IN



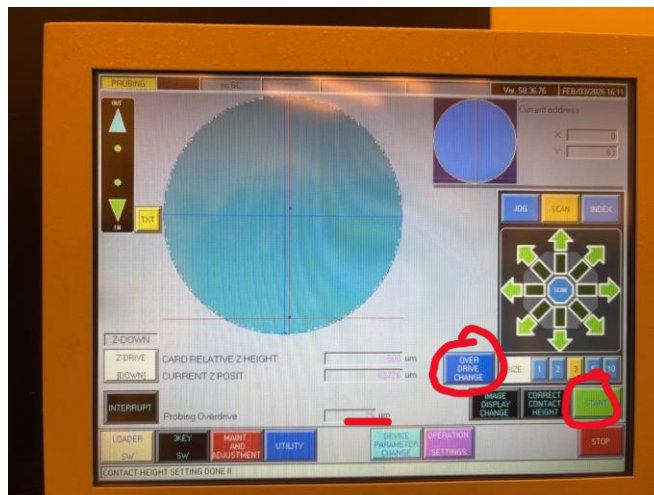
14. If there is another prompt to select UNIQUE POINT, make sure that cross hairs are on top left inner corner of product die and hit DATA IN. Once you have confirmed cross hair location, select ANGLE ADJUST END.



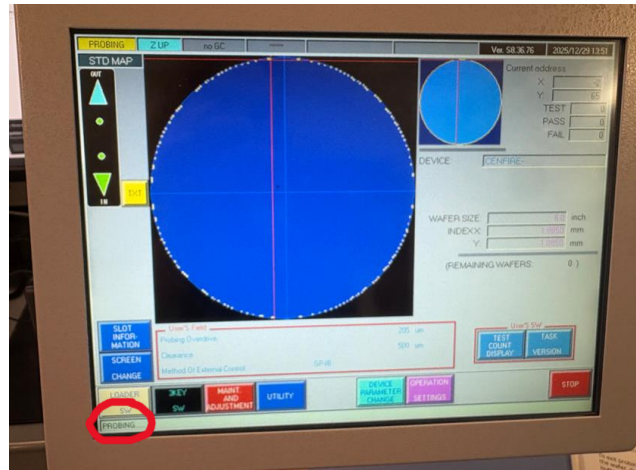
15. If you get a prompt to Stop after auto needle height setting done, clear screen



16. Edit Probing Overdrive if needed (default 25um). Then hit START.



17. Now the probe should be in "Probing" mode

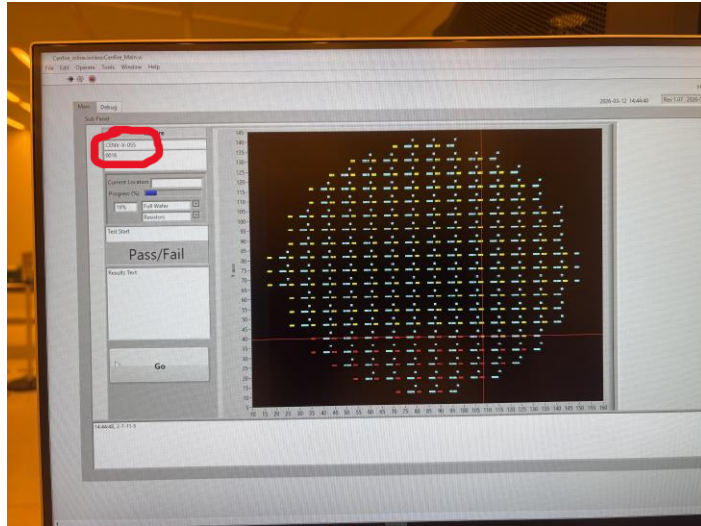


7. Start Test

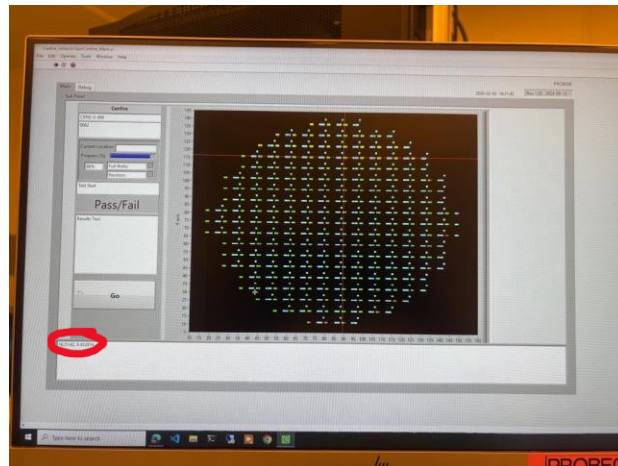
1. On the PC, open the CENFIRE-INLINE program if it is not already open from previous wafer



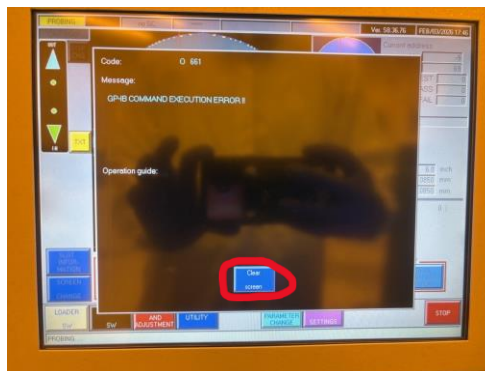
2. Enter the Via wafer Lot and wafer ID in the user prompts. If wafer in lot already ran, simply change the wafer ID. Once data is entered, hit Go
(EXAMPLE DATA ENTRY: Lot ID: CENV-V-0##, Wafer ID: 00##)



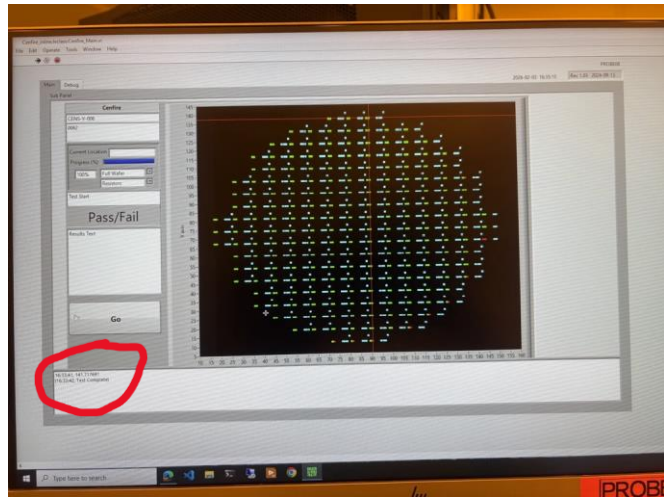
3. While test is running, monitor resistance measurements in bottom window. They should alternate between about 0.5 Ohms and 150 Ohms.



4. While Test is running, clear any GPIB warnings on prober display if desired

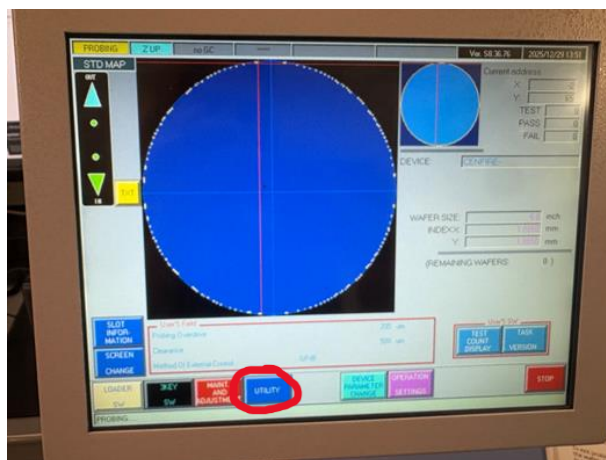


5. Watch for Test Complete. Data is automatically uploaded to Cenfire database.

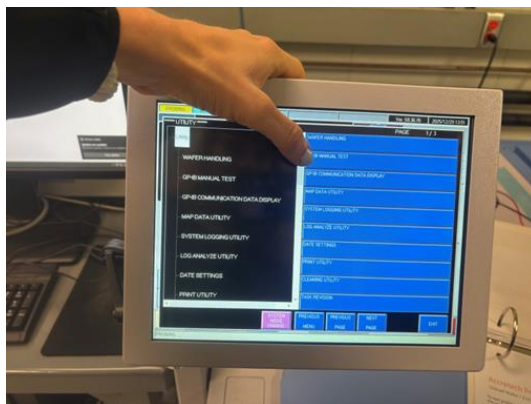


1. Unload the Wafer

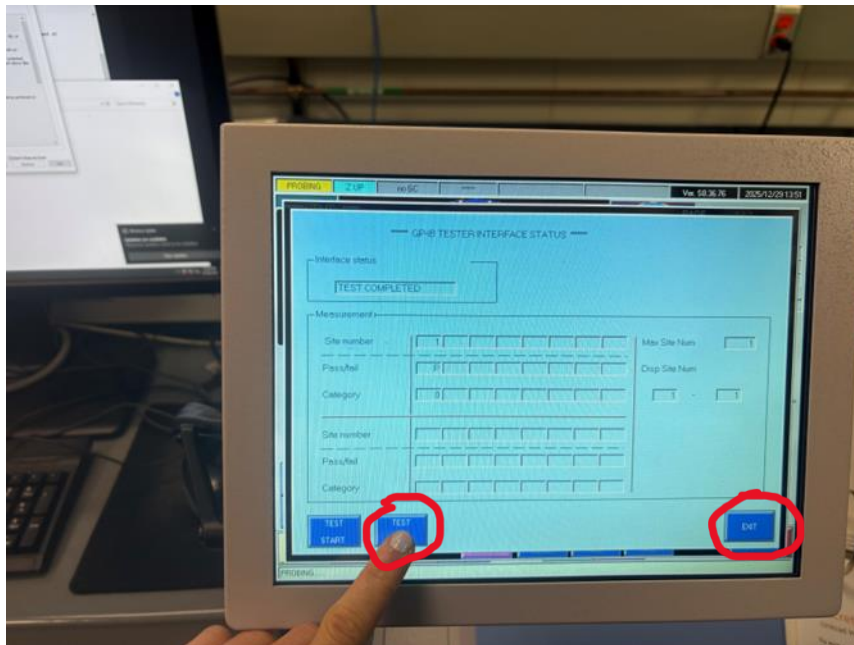
1. Select UTILITY



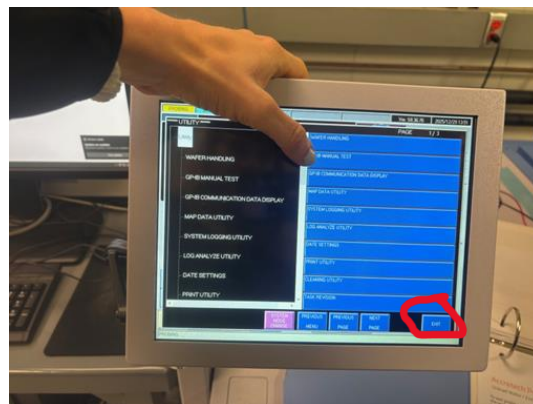
2. Select GPIB MANUAL TEST



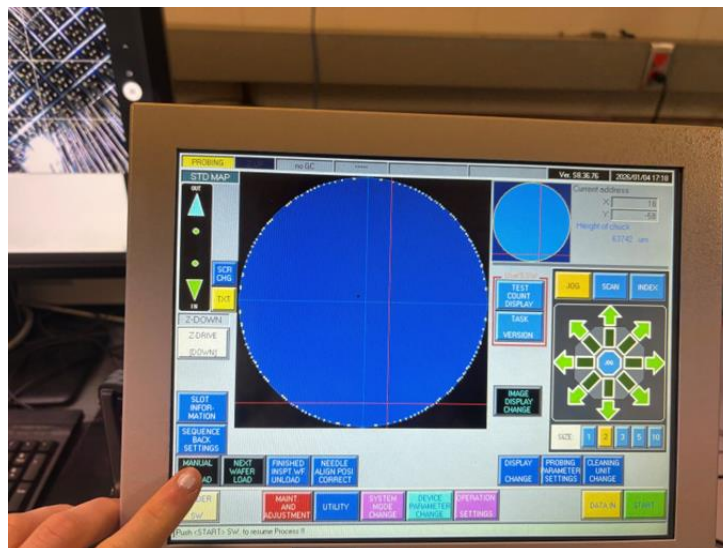
3. Select TEST END twice, then EXIT



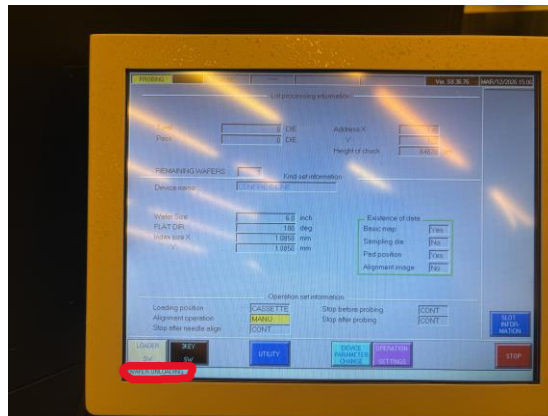
4. Select EXIT



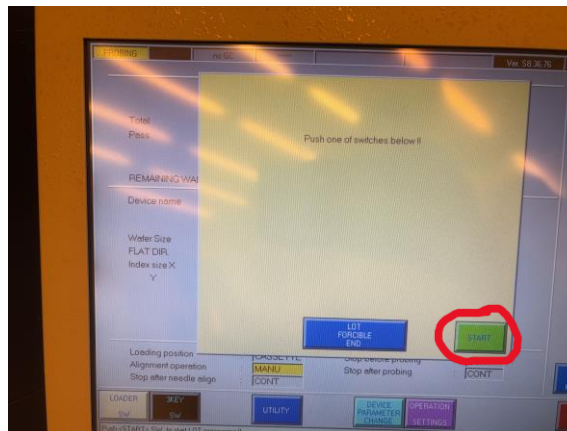
5. Select MANUAL UNLOAD 2x (status will say “WAFER UNLOADING”)



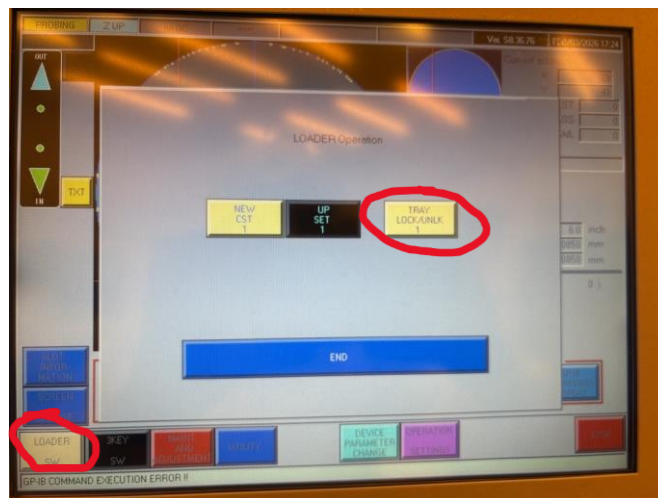
6. Wait for wafer to unload (status says “Wafer Unloading”)



7. If there are more than one wafer in the cassette, when this pop-up appears, select START to load the next wafer



8. Once wafer has unloaded, select LOADER SW and UNLOCK/LOCK to unlock loader and retrieve the wafers from the cassette



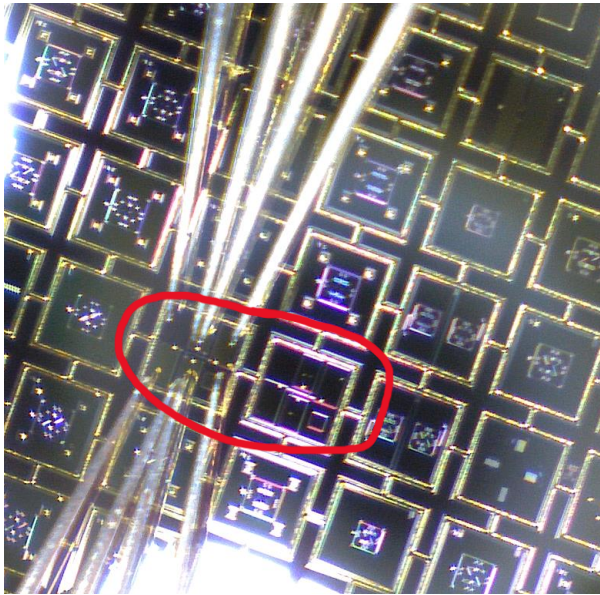
Appendix:

Validation

Use CENV-V-049 to validate tests (Resistance values should be between 0.5 and 150 Ohms).

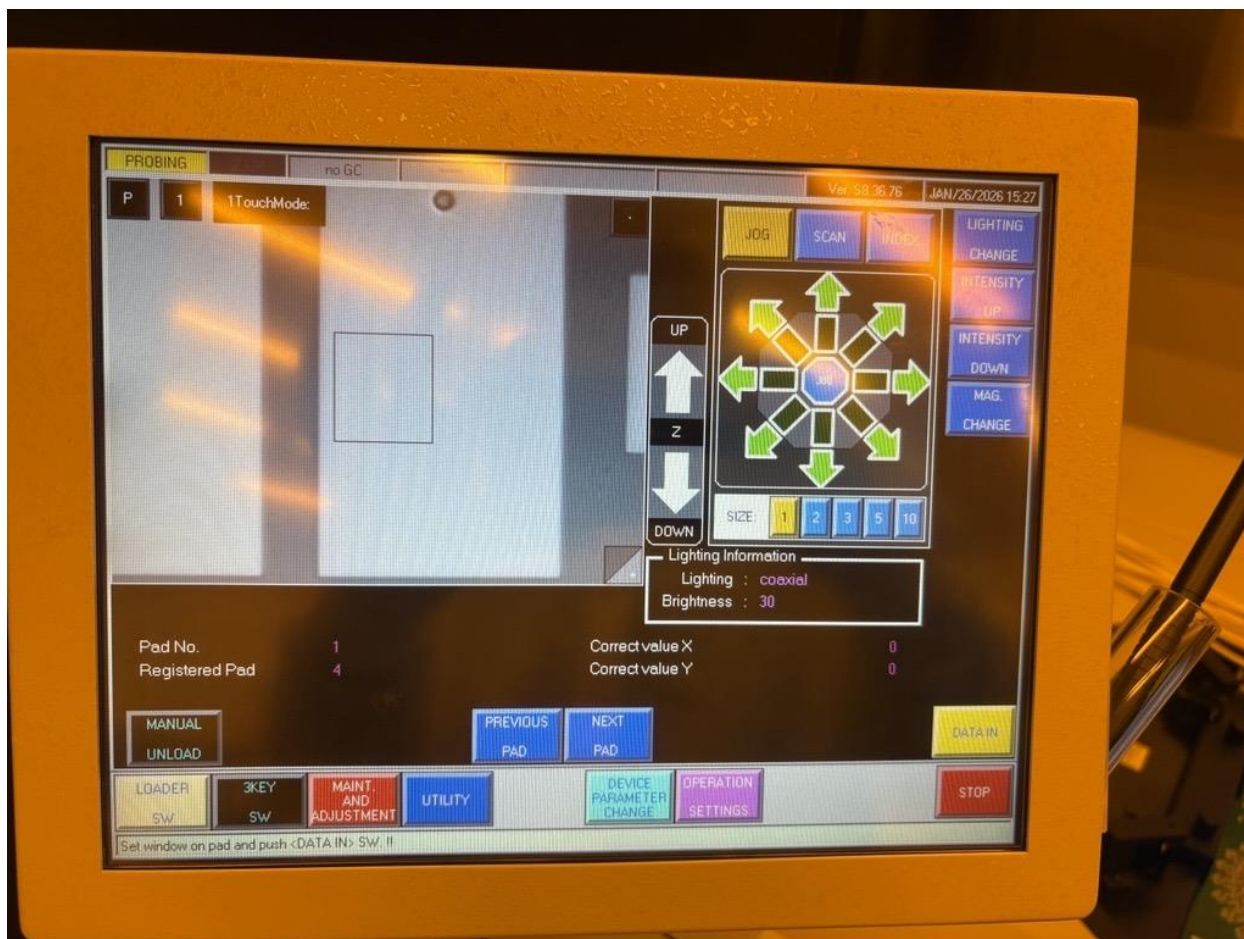
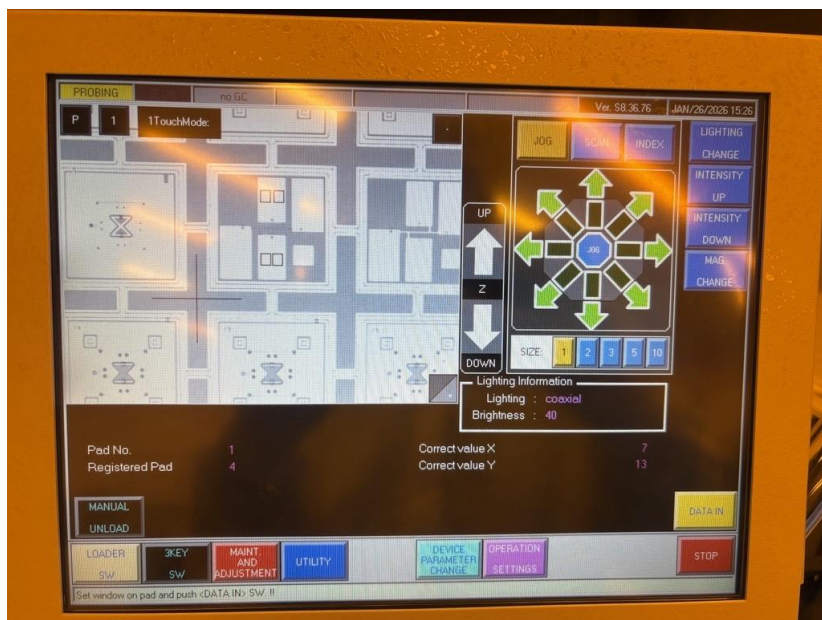
Purpose of Test:

4-wire kelvin resistance measurement of test structure



Defining pads:

During product file creation, the pads should be defined using the resistance test structure as shown in the image. Because the pads are large, the spacing of the pads defined (relatively arbitrary when referencing the test structure) may not exactly line up with needle tips. This may take some experimentation to optimize the pad definition with the needle tip spacing. Also, this unprecise pad definition causes needle find function to always fail, and have to be done manually. This die that the pads are defined on must be referenced for the Needle Height Adjustment Die (absolute coordinates).



Needle Height Adjustment Die:

Needle alignment die setting (a parameter in DEVICE PARAMETER SETTINGS --> Needle Alignment Data Settings) is set to the test structure with respect to the reference die. This die must be set accurately, or else the height of the needle adjustment die will be too far off from the die it was trained (when pads were defined). This will cause a software Z out of bounds error during wafer loading.

Absolute Coordinates

X = 0

Y = 63

During manual alignment, the SET WINDOW ON PAD step must be done with the product die centered during this alignment step (do not scan to the test structure, this will cause a software XY out of bounds error during loading).

Reference Die:

The reference die (a parameter in DEVICE PARAMETER SETTINGS --> Reference Die Settings) is set to:

Teaching

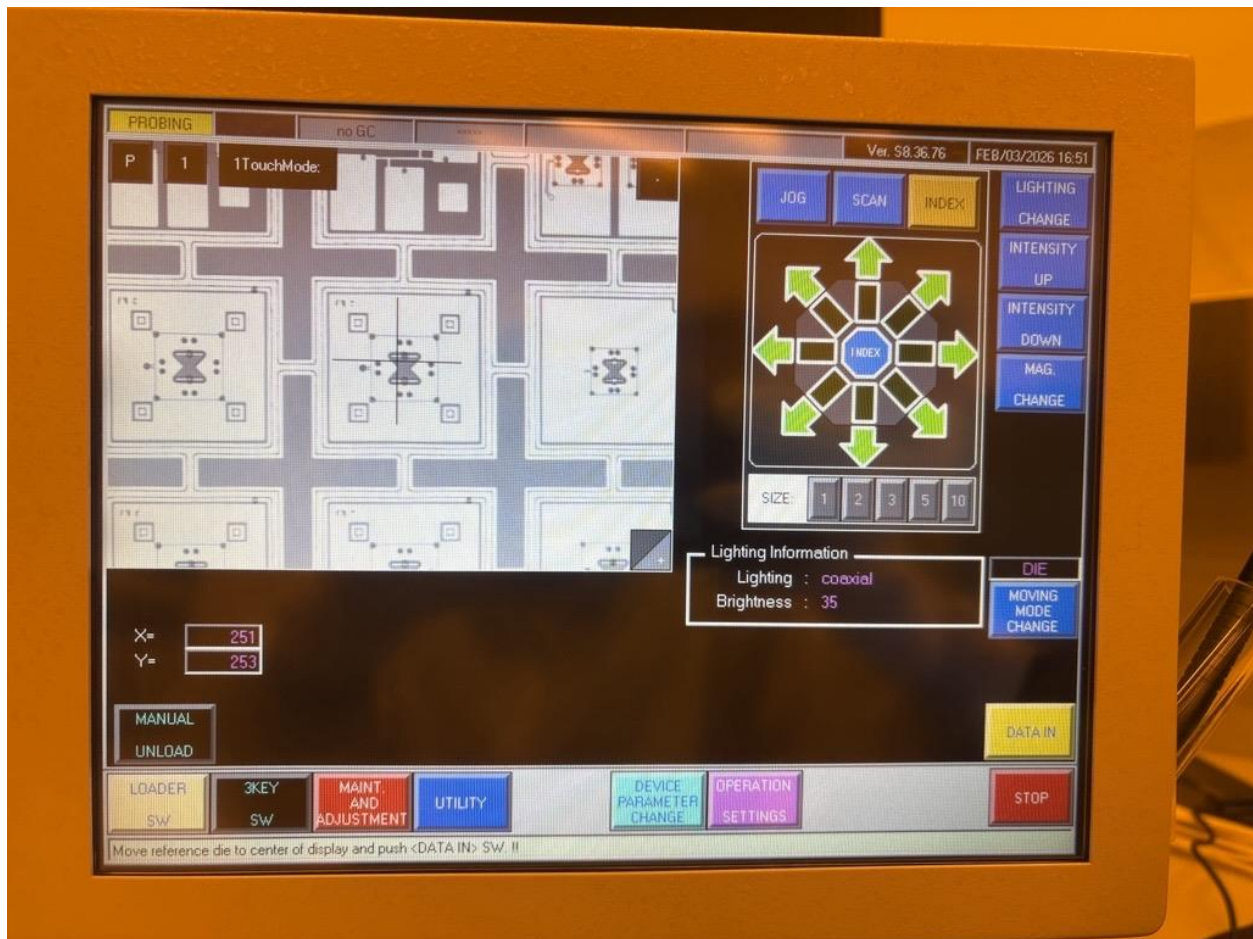
X = 0

Y = 0

When training the prober on the reference die, a new device must be made, and the alignment data must not be copied to the new device. When defining the reference die, the absolute coordinates are:

X = 251

Y = 253



Finding Z Height and overdrive ('First Touch') Procedures:

When get to Contact Height adjustment page – alternate between adjusting Card Relative Probe Height (height of chuck with respect to needles) and "IMAGE DISPLAY CHANGE" (camera) to look for contact marks (or from probing mode, select "Correct Contact Height" from overdrive edit display). Set the die you are contact with the wafer map "index" arrows. When hit IMAGE DISPLAY CHANGE test die will be centered).

CARD RELATIVE Z HEIGHT starts at -500um

Set CARD RELATIVE Z HEIGHT to about -350 to start, this moves chuck up

Select Z-DRIVE [UP] to add -500um (move chuck back down) before hitting IMAGE DISPLAY CHANGE

IMAGE DISPLAY CHANGE will show you where the needles were under the camera. Look for probe marks

Iterate through moving the chuck up (Reduce from -350, to -200, to -150, etc... should contact at 0um. May need to go positive um) until you see probe marks

ALWAYS HIT Z-DRIVE [UP] BEFORE IMAGE DISPLAYCHANGE (move chuck down before it moves over to the camera)

Once faint contact marks are visible, now add overdrive to see the "scratch"

if probe marks appear to be in wrong spot, use "NEEDLE ALIGN POSI CORRECT" (if you are over correction limit, you cannot fix this)

- this tells you difference from where it thinks probes are landing (where you lined up needle crosshairs and pad definition) and where it actually is landing

-if probe marks are off, this is an indication that probe card is damaged

